# Department of Computer Engineering
# Faculty of Engineering, University of Peradeniya

# CO221 : Digital Design
# Project: Building digital modules of a simple computer

In this project your task is to build modules of a simple computer system. To build modules in this project, the knowledge of Digital Design Course is sufficient. Though building computer systems is covered by the Computer Architecture course the knowledge of Computer Architecture is not required for this project. However, by doing the project, in addition to learning on developing digital modules, this project introduces concepts of a simple computer and hence it will help you to follow computer architecture course in the coming semester.

The project is based on SAP-1 Computer described from page 140 of Malvino and Brown book. An ebook is attached.

Description of the project is given below

As show in in Fig1, simple computer has following modules

1. Register A (Accumulator)
2. Register B
3. Arithmetic and Logic Unit (ALU)
4. Program Counter (PC)
5. Instruction Register (IR)
6. Memory Address Register (MAR)
7. Random Access Memory (RAM)
8. Output Register
9. Controller/Sequencer

These modules are interconnected via 8 bit data bus

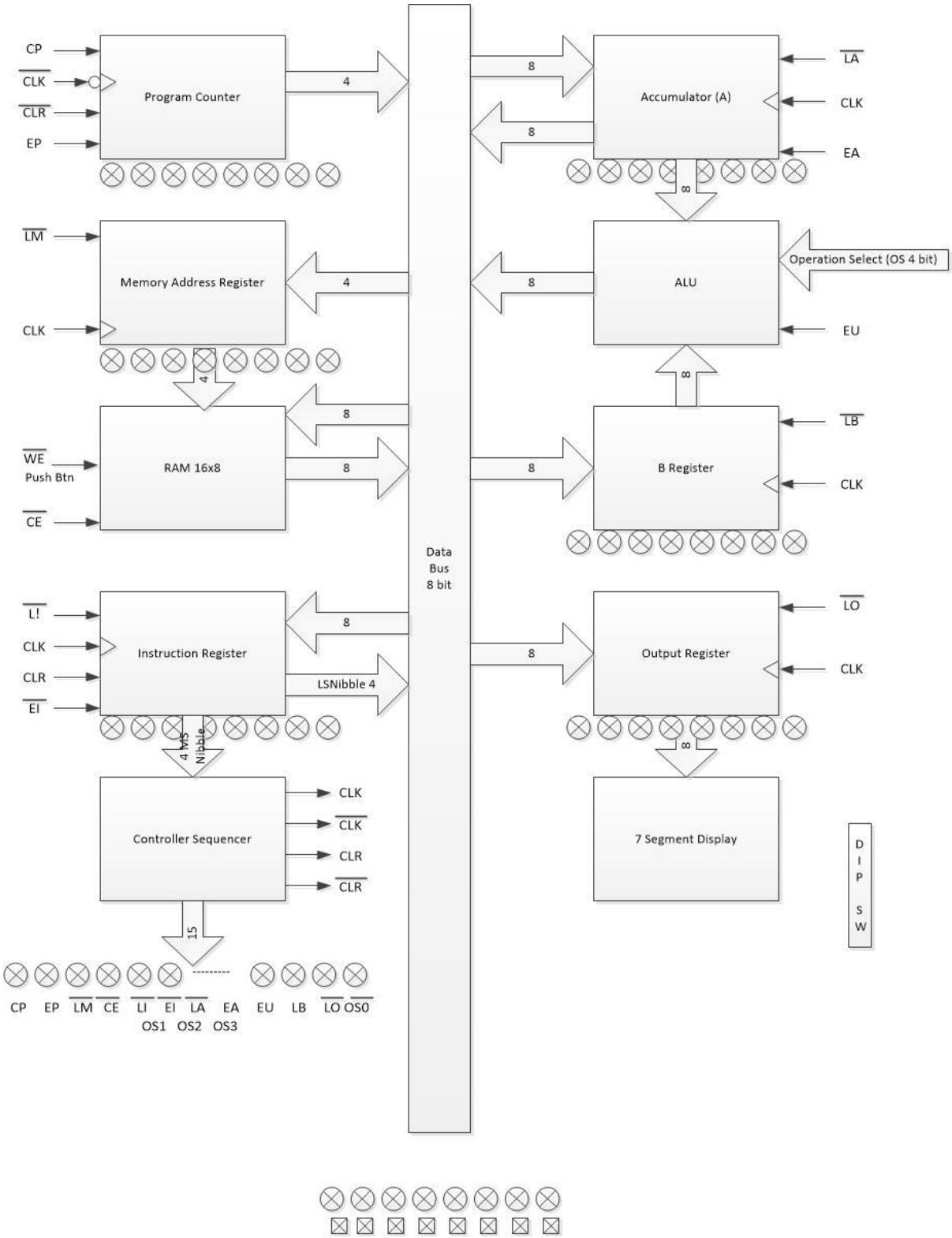


Fig 1, Block diagram of a simple computer

The operation of each module is given below

**Register A**

Load 8 bit from the bus when La is low with the CLK
Output 8 bit to bus when Ea is high with the CLK, when Ea is low o/p is at high impedance.
Always output the stored value to the second 8bit output (connected to ALU)

**Register B**
Load 8 bit from the bus when Lb is low with the CLK
Always output the stored value to the 8bit output (connected to ALU)

**Program Counter**
4 bit counter start from 0000 and increment 1 bit at a time when Cp is high with the –CLK. Output 4 bit to W bus when Ep is high with the CLK. It has active low direct clear input –CLR.

**Arithmetic and Logic Unit (ALU)**
ALU has 3 bit operation selector inputs, two 8bit inputs and one 8bit output. No clock is used. A block diagram of ALU is shown in Fig 2.

Arithmetic modules

Inside of ALU each of following arithmetic operations is implemented as independent arithmetic modules using basic logic gates. The two 8 bit inputs are always connected to an arithmetic modules and output of each module is always available. The multiplexer unit selects the appropriate output.

| Operation | O1 | O2 | O3 |
|---|---|---|---|
| Addition | 0 | 0 | 0 |
| Subtraction | 0 | 0 | 1 |
| Bitwise XOR | 0 | 1 | 0 |
| Bitwise AND | 0 | 1 | 1 |
| Bitwise OR | 1 | 0 | 0 |
| Bitwise NAND | 1 | 0 | 1 |
| Bitwise NOR | 1 | 1 | 0 |

Table 1. *Operation Select* input

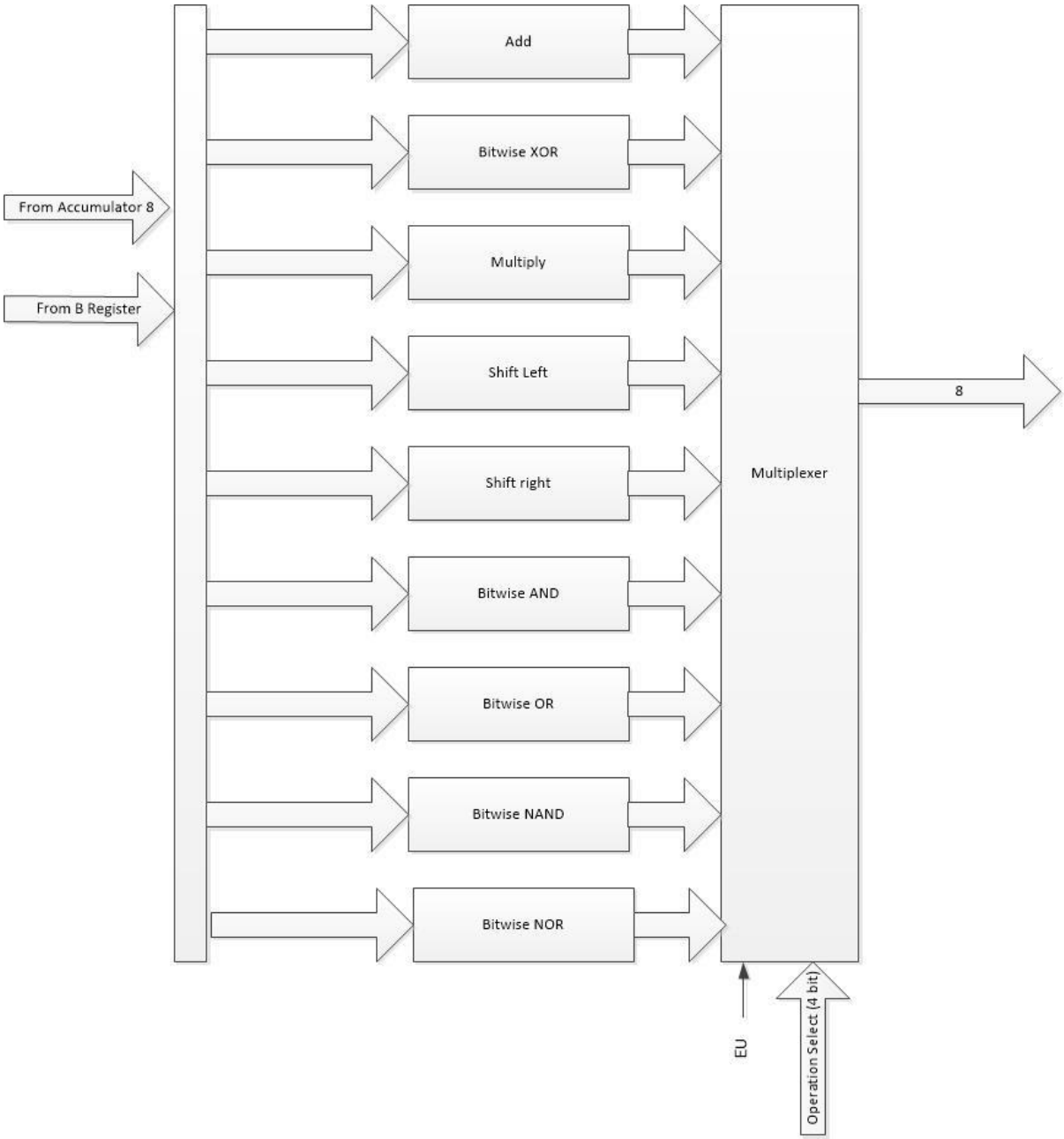


Fig 2. A block diagram of ALU

**Multiplexer unit of ALU**

Multiplexer unit has 8 bit input and 4 bit *Operation Select* input. The output of all arithmetic modules are connected to this multiplexer module and one output of arithmetic modules will be given as the output of ALU based on the select input bit combination.

When arithmetic and logical operation is to be performed, first the program load two operands to A and B register and set the *Operation Select* input corresponding to the mathematical operation. The EU signal will send the ALU output on to the bus.

Table 1 gives *Operation Select* values you must use when selecting each operation.

## Controller Sequencer Modules

Block diagram of a controller sequencer is given in Fig 3. For a given command opcode, correct sequence of control actions (Micro instruction) to be given to other modules should be generated at each active output of ring counter (Active high signal of T1 to T6).

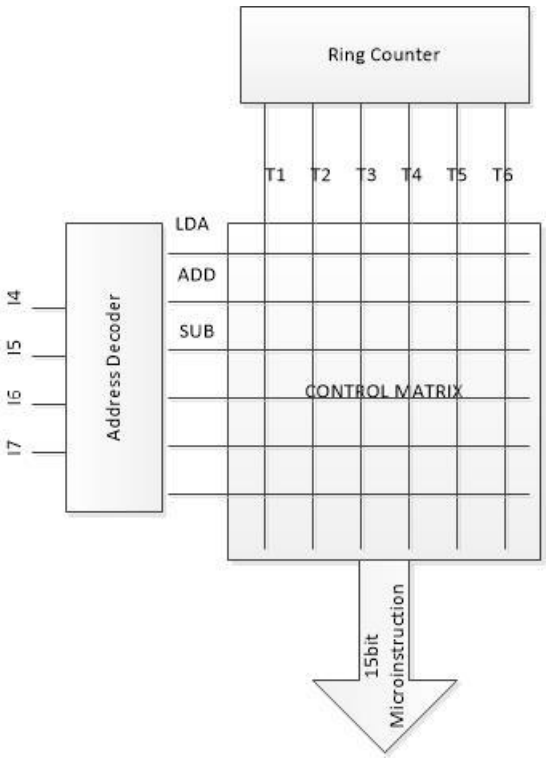


Fig 3. Controller Sequencer

For example for LDA XH command, following actions should be carried out.

| Ring counter output | Action | Activate commands |
|---|---|---|
| T1 | Transfer address of the PC to MAR | EP, -LM |
| T2 | Increment program counter | CP |
| T3 | Transfer instruction from memory to IR | -CE,-LI |
| T4 | Transfer address XH to MAR | -EI, -LM |
| T5 | MAR addressed data in the RAM is loaded to accumulator | -CE, -LA |
| T6 | No operaton | None |

Table 2. Controller sequence for LDA command

The modules that gives outputs to the data bus should be Tri-state so that only selected modules will send the output to the data bus and remaining outputs are at high-impedance state.

A set of LEDs should be available to demonstrate the operation of each module.

**Project implementation**

You are free to form Groups of 3 or 4 and free to select any activity. Please let me know the activity that each group has selected within the next week. Each group should make a PCB. A computer will be made by connecting all PCB together. The difficulties of modules are not equal. So please form group of four if the module is complex.

The modules should be implemented using basic combinatorial logic circuits of 74 series ICs available with you or in the lab. Even though the complete circuit seems to be complicated, a module developed by each group is simple.

Phase 1: Implement modules using Proteus.

You are required to submit the computer simulation (a single zip file) within a week to the link in FEeLs.

Phase 2: Fabricate PCBs and implement modules

Phase 3: Combine PCBs to make the computer.

| Activity | Module to be implemented |
|---|---|
| 1 | Program counter |
| 2 | Accumulator |
| 3 | Memory Address Register |
| 4 | RAM |
| 5 | Instruction Register |
| 6 | B Register |
| 7 | Output register |
| 8 | 7 Segment Display |
| 9 | ALU-Add |
| 10 | ALU-Sub |
| 11 | ALU-XOR |
| 12 | ALU-AND |
| 13 | ALU-OR |
| 14 | ALU-NAND |
| 15 | ALU-NOR |
| 16 | Multiplexer |
| 17 | Six bit Ring counter |

| | |
|---|---|
| 18 | Address Decoder |
| 19 | Control Matrix-Opcode LDA: Load RAM data into accumulator |
| 20 | Control Matrix-Opcode Add: Add RAM data to accumulator |
| 21 | Control Matrix –Opcode SUB Subtract RAM data from accumulator |
| 22 | Control Matrix –Opcode OUT Load accumulator data into output register |
| 23 | Control Matrix –Opcode HLT Stop Processing |
| 24 | Control Matrix -Shift Left |
| 25 | Control Matrix -Shift Right |
| 26 | Control Matrix –AND |
| 27 | Control Matrix –OR |

Table 3. Assignments of modules to groups

At the end of the project each group will be interviewed. The examiner will ask anyone in the group to explain a given section of the module. All members of the group will be asked to explain a part.

If you have any issue please contact Dr. Kamalanath Samarakoon immediately.